

Specialist in the innovation of low-statistics modeling, time-series analysis, and the reduction of complex data. I extract weak signals by designing and deploying robust, reproducible analysis pipelines while maintaining rigorous version control. I consistently deliver high-impact results through advanced analysis and scripting in Python, Bash, and C, and excel at communication of complex ideas with creative visualizations custom built in Blender and Xcode. Due to my expertise in coding, I am an extremely versatile asset in analysis and modeling not just in astrophysics, but of data in all STEM fields.

EXPERIENCE

Lead Scientist (5+ years) - Developed, maintained, and published custom analysis pipelines (python, bash, C; >3k lines) to model complex physical systems and extract signals from limited and noisy data. - Led and coordinated multiple research teams to deliver high-quality scientific products and ensure timely (~ 6 – 12 months) project completion. - Produced results over 30× more informative than competing analyses, advancing active research areas and publishing in a leading peer-reviewed physics journal.	2019 — Present <i>The George Washington University Department of Physics, Washington D.C., USA</i>
Scientist (5+ years) - Conducted time series analysis (python; ~1k lines), spatial analysis (python; ~1k lines), and modeling of low-statistics data (python, C, bash; >2k lines), improving current models by 10%. - Produced additional high-impact results published in leading peer-reviewed physics journals. - Successfully secured funding from competitive, over-subscribed (8x) NASA research solicitations (~ \$20, 000 USD).	2019 — Present <i>UMBC/CRESST II NASA GSFC Greenbelt, MD, USA</i>
Lead ICT Math/Physics Teacher - Taught Physics, Geometry, and Algebra to students requiring Integrated Co-Teaching to succeed, leading to a 100% pass rate for state-wide testing. - Constructed teaching plans for students, and planned short- and long-term curriculae for each academic year and reduced deliverables by 50%.	2017 – 2019 <i>Success Academy Charter School New York City, NY, USA</i>

TECHNICAL SKILLS

Tools and Languages	Python (Numpy, Matplotlib, SciPy, Pandas), Git/Github, $\LaTeX$, R, HPC, MATLAB, C/C++, PERL, AWS, SQL, Machine Learning Packages (TensorFlow, PyTorch), Monte-Carlo Markov Chain
Areas of Expertise	Mathematical optimization, Mathematical Modeling, Bayesian Statistical Analysis, Time-Series and Fourier Analysis, Spatial Mapping, Data Cleaning, Reduction and Pipeline Generation, Visualization and Presentation, Oral Speaking

EDUCATION

Doctorate of Philosophy in Physics , <i>The George Washington University</i>	2021 - 2025
<i>The George Washington Dissertation Completion Fellowship</i>	2025
<i>Bell Atlantic and an Andrew Knox Endowment Fellowship</i>	2024–2025
<i>Bennhold Family Award for Best Teaching on the Part of a Graduate Student</i>	2020
Master of Philosophy in Physics , <i>The George Washington University</i>	2021 - 2023
Certificate of Data Science , <i>The George Washington University</i>	2021 - 2023
Master of Science in Physics , <i>The George Washington University</i>	2019 - 2021
Bachelor of Arts in Physics and Astronomy , <i>Vassar College</i>	2013 - 2017

FIRST-AUTHOR PUBLICATIONS (CLICK FOR FULL ORCID RECORD)

[1] Lange, A., et al., “The Vela Pulsar and Its Pulsar Wind Nebula Vela-X Using 13 Yr of Fermi-LAT Observations”, *The Astrophysical Journal*, vol. 988, no. 2, Art. no. 200, IOP, 2025. doi:[10.3847/1538-4357/ade713](https://doi.org/10.3847/1538-4357/ade713).
[2] Lange, A., et al., “Orbital Phase-resolved Analysis of X-Ray and Gamma-Ray Observations of the High-mass Gamma-Ray Binary 4FGL J1405.1-6119”, *The Astrophysical Journal*, vol. 986, no. 2, Art. no. 181, IOP, 2025. doi:[10.3847/1538-4357/adda32](https://doi.org/10.3847/1538-4357/adda32).
[3] Lange, A., “FermiPhased: A tool for phase-resolved likelihood analysis of Fermi-LAT data.”, *Arxiv*. arxiv.org/pdf/2511.04810.

ACTIVITIES

Led, organized and hosted the annual DC-area astronomy and astrophysics conference.	2022 — 2025
Graduate Teaching Assistant of Physics, Astronomy and Mechanical and Aerospace Engineering	2019 — 2023
News Publication of DC-Area Astrophysics Graduate Student Conference Revived!. American Astronomy Society, Education Committee Blog	2024
Featured Oral Presentation at the DIAS-CDY Workshop - Gamma-Ray Loud Binaries. Dublin, Ireland	2024
Featured Oral Presentation at the 11th International Fermi Symposium. College Park, University of Maryland	2024